Ensemble Learning with Exploratory Data Analysis for Predicting Covid-19 Epidemic Trends

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Abstract

The COVID-19 Coronavirus, namely SARS-CoV-2, has sent the world into disarray ever since it started spreading globally, originating from China. In an attempt to combat this pandemic, our research has led us to Machine Learning (ML) based forecasting mechanisms. As such, we've used the following ML models in our research: Holt's model using Exponential Smoothing, and Support Vector Regressor (SVR), as we tried to hone in on the deadly factors of COVID-19. Four primary factors were focused on: number of new cases, number of deaths, number of patients recovered and number of active cases. Upon rigorous testing, it was found that the Ensemble model was the most effective, and that the Holt's Winter Model and Holt's Linear Model performed quite impressively as well. SVR proved to be the most ineffective throughout all the factors. After acquiring the data from three different sources, we visualised the data through Exploratory Data Analysis. It was found that the population-density of a country did not play a large role in the number of COVID-19 cases. India is an example of a country that had the highest number of recovered cases and a relatively low number of confirmed and death cases Furthermore, more cases ended in fatalities within low-density populations, which could be seen in the case of the US, recording the highest in terms of confirmed cases and death cases. The information was merged, and then polished and cleaned via pre-processing and then was fitted into the model to achieve our results. As such, this study could help in the fight against COVID-19 and future similar cases, with more efficiency.

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Chapter 1

Introduction

COVID-19 Coronavirus, namely known as SARS-Cov-2, has become the talk of the world ever since it has started taking over. We have had very dangerous viruses present in our ecosystem for a long time, but this suddenly appearing virus has demolished the global routine in every sector, and left the world at a standstill for quite some time. As time has passed, researchers, scientists and doctors among others, have worked diligently to find out how to live safely in the new normal. COVID-19 has left its mark on the world after originating from Wuhan, China in 2019, with an approximated 450 thousand deaths as of June 2020. This epidemic quickly turned into a pandemic, due to the high level of contagiousness it possesses, and the ease through which it spreads. [1] Thus, high level steps had been taken globally in order to ensure this virus was contained. This included the addition of social distancing and quarantining of countries worldwide. Since the virus was found to be spread through bodily fluids when entering through the mouth, the usage of masks was also highly recommended. More and more medical studies were and are being conducted in order to uncover all the information on this virus. [2, 3, 4, 5] Along with these studies, there were also studies taking place that honed in on all the various economic, social and psychological changes that would occur in societies across the world.[6, 7, 8, 9, 10] In order to be able to forecast the various effects on this virus and its foundation, there was also an introduction of statistical, mathematical and epidemiological models.[11, 12, 13] Furthermore, the effectiveness of Machine Learning (ML) models and approaches were also delved into, in order to deal with it better.[14, 15, 16, 17] With how unprecedented this virus has become in modern society; all these studies have become a necessity in or-

der to help stave off this virus as soon as possible [18]. The use of Machine Learning (ML) algorithms and models, and other artificial intelligence methods have become quite popular in modern times, and they are being used in various fields.[19] When using ML in these applications, various domain fields can be used such as healthcare, business applications, intelligent robots, gaming, autonomous vehicle (AV), modeling, image processing, natural language processing and voice. These algorithms usually follow a trial-and-error method based on decision statements such as if-else.[20] With various types of ML algorithms having been used for this purpose, they have been known to be the best for including disease forecasting, weather forecasting among various other needs. They are also useful in disease prognosis, which can be bettered in the future to help patients with very specific needs and diseases. [21] A lot of studies have also been conducted with ML methods for diseases such as coronary artery disease, cardiovascular disease prediction and breast cancer prediction.[22] A very important part is that this study is focused on [23] how significant these methods can be in order to get live updates on COVID-19 cases, and can help in getting responses in on time. [24] Thus, this will allow everyone to be able to deal with this pandemic in a much more prepared manner. The main point of this study is to be able to provide a forecast model of the spread of COVID-19. [13] Originating from Wuhan China, this virus is known to develop pneumonia-like symptoms in the victim's body [25]. However, it is not restricted to this behavior, and can be different depending on the people. Some suffer from severe acute respiratory syndrome and multi-organ failure, which can be fatal. [26] With millions of people having been affected throughout the world since the virus originated in late 2019, it brought about attention from researchers and scientists all over the world. It is spreading mostly by respiratory droplets, handling contaminated objects or surfaces and by contact between people. One of the most problematic features of this virus is that it can remain unnoticed for a few days, as it may not show any symptoms initially. This made the need for lock-downs and quarantines quite significant throughout the world, especially with the lack of a universal vaccine. The search for vaccines for this epidemic began immediately, and a few different vaccines have been produced till date, which appear to be promising. However, the danger is still looming large due to the virus continuously mutating and creating newer strains throughout the world. England and South Africa are just two of the places which have produced new strains. Since it is such a massive issue globally, our study aims to develop a

COVID-19 forecasting system. We mainly take into account four variables, and they include; number of newly confirmed cases, number of deaths, number of recovered patients and number of active cases. The ML algorithms used in this study, namely; the Polynomial Regression, the Holt's Linear, the Holt's Winter and the Support Vector Regressor (SVR), have all been trained with the datasets provided by John Hopkins, Kaggle, World Health Organizations (WHO), alongside the Ensemble model. To clean the data, it has also been pre-processed and then divided into three subsets, which were used in the performance evaluation. With this pandemic being quite an unexpected turn of events for the entire world, and having turned the lives of billions upside down, we feel that it is quite important to learn the nature of such events, specifically the ongoing one. Thus, we feel the need to delve into the various avenues affected by COVID-19 and see how it has impacted us in our lives till this date, and in the ongoing future.

1.1 Project Overview

COVID-19 has taken the world by a storm and had brought it to standstill for quite some time, with everyone around the world having to double-check their approach to life. From late 2019 till the current date, we are stuck with new rulings to help fight this virus, and as such, it would be quite useful to be able to forecast its spread and effects, which is where ML comes in. ML forecasting has increased its value over the years, with it being applicable in numerous sectors of the economy, due to the availability of the numerous ML algorithms. Apart from COVID-19, machine learning has had its applications in providing regression and neural network models, which help in the study and prognosis of patients. This also includes Machine Learning's effectiveness when it comes to the prediction of fatal diseases of the heart, or cancer spreading throughout the body. ML can also help with the monitoring of the economy via the stock market as well. Overall, ML has its many functions especially in this modern age, and we have applied it to have a live forecast of the COVID-19 epidemic's confirmed cases. With the need for interventions being largely necessary, and each choice proving to be very heavy, it would be infinitely helpful to those in charge, to know the proper decisions to be made; which is where Machine Learning comes into play. Thus, this study is aiming to achieve an early forecast model for the officially named novel coronavirus: SARS CoV-2, by the World Health Organization (WHO). We have designed a framework where, we will do problem identification

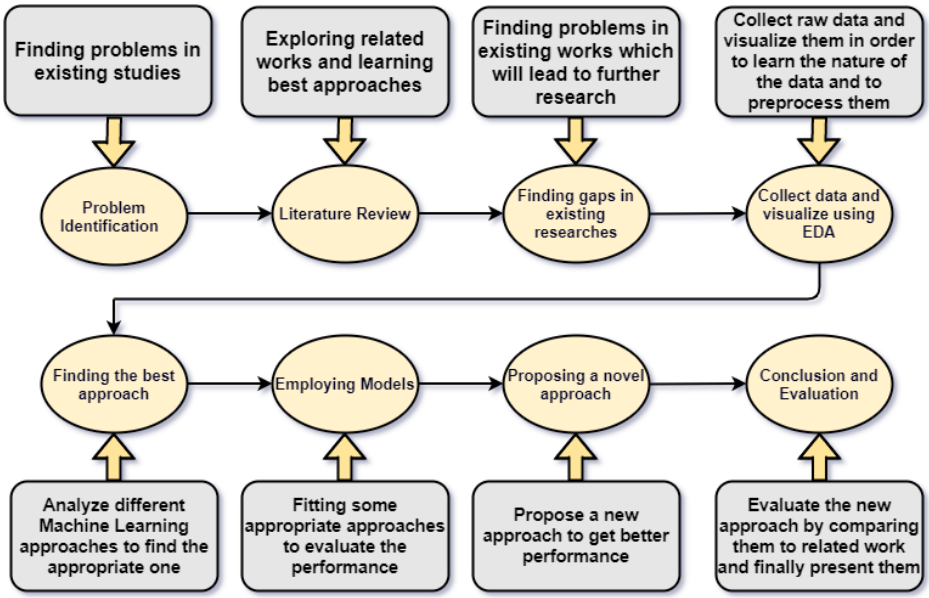


Figure 1.1: Project's Framework

on previous work, then do some literature review. then analyze the different ML approaches.The project framework can be shown in **Figure 1.1**.

1.2 Motivation

Last year, the world was shaken by the novel coronavirus and its consequences. Almost all the countries and regions of this planet were affected by the virus and still, the human race is going through a hard time combating it. However, the spread of the virus in all regions and the effect of the virus on different types of people are not the same. It's been seen that by taking different effective measures, the spread was lessened. For example, as of October 14, 2020, the spread was on the rise in Italy until they imposed a regional lockdown across the country on November 6, 2020. After that, the spread was decreasing significantly till December 23, 2020. Moreover, some people survived the disease after taking some or even no treatment while some couldn't do so. And the higher death rate of older people and people with some chronic diseases supports that claim. So it's quite clear that the consequences, the level of contamination and the spread of the virus are dependent on different features. So, it is very crucial to predict

the trend of the virus' spread in different situations and to extract the critical measures taken in the past i.e. the features from previous data that changed the level of the spread so that in the future, useful and effective precautions can be taken by the government to reduce the level of loss and the spread of the virus and avoid dangerous consequences. These motivated us much in doing the project.

1.3 Objectives

The objectives that are required to meet the desired goal of this project are as follows:

- To get some knowledge of the topic through literature review on COVID-19 and ML.
- To collect updated data related to the COVID-19 coronavirus from reliable sources.
- To visualize the data using different Exploratory Data Analysis (EDA) techniques to learn some pattern from the data.
- To select the ML model(s) to be built based on the information extracted by data visualization.
- To pre-process the data based on the information extracted in data visualization part.
- To fit the pre-processed data in the selected ML model(s) to extract critical features from the data and predict the drift of the spread of the virus.

1.4 Methodology

The aim of our study is to be able to provide an accurate forecasting of COVID-19 and its spread throughout the world, focusing mainly on the number of newly confirmed cases, number of recovered patients and the number of death cases. The dataset used for this study was acquired from the Github repository provided by the Center for Systems Science and Engineering, John Hopkins University, and Kaggle [27]. The repository contains the time series dataset which includes the number of all the date-wise confirmed cases, recovery cases, and

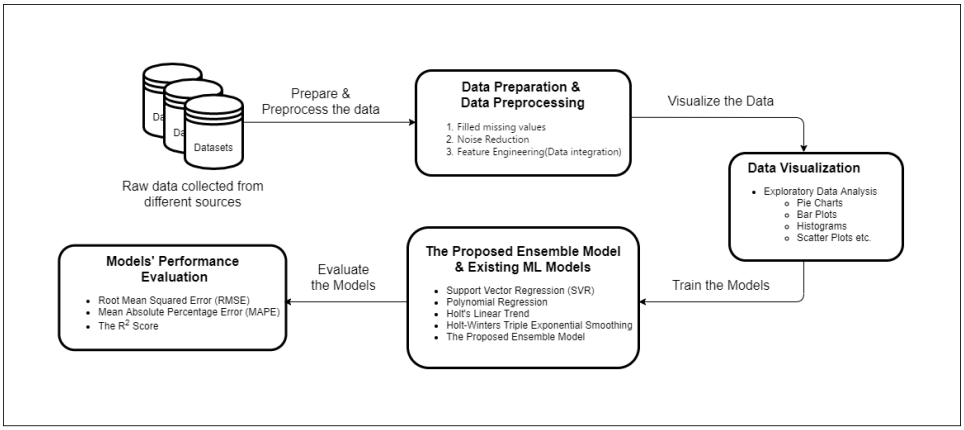


Figure 1.2: Project methodology in brief

death cases. All this data is also updated everyday. This study is conducted for COVID-19 predictions, which is a novel coronavirus that has plagued the world from late 2019. Four different ML models were used in this study in order to find the best possible one; Support Vector Regression (SVR), Holt-Winters Triple Exponential Smoothing, Polynomial Regression and Holt's Linear Trend. Since the models did not provide desirable results, an ensemble model was developed with an amalgamation of all of them, with the weighted average. For the study, we first collected all the necessary raw data. The data was visualized by identifying noise, extracting patterns and identifying the nature in them. It was also pre-processed through Feature Extraction and Data Cleaning. This is repeated for as long as needed. Then, the data was fit into a train model after being pre-processed and based on its nature, for model selection. After this, the model evaluation was done and if it was a good performance, it was concluded, based on metrics such as Root Mean Square Error (RMSE), etc. If the data was not desirable, the process was repeated. The methodology is shown in **Figure 1.2**.

1.5 Project Outcome

With the research into ML algorithms showing their significant effectiveness when it comes to predictive forecasting for COVID-19 cases, the human race's attempts in fighting these situations will increase dramatically in future cases, and our survival rate will be better for this pandemic. This is why we have taken upon this project, that helps us be able to use and optimize these models for

the betterment of the human race. With proper data extrapolation and visualization, the ML models will allow us to understand the features of this virus and how it is affecting us, which will then let us fight it better. Not to mention, it will prepare us to fight these situations at a much faster rate in the future. This will benefit the overall population of humankind. If we are able to accomplish our goal and have our project be accepted, it will also enable us to publish our paper, which will inspire us to dig deeper into these models and any other means to help mankind.

1.6 Organization of the Report

This paper is split into six chapters containing all the information about the project's motivation, working methods, requirement analysis, results, and concluding statements.

- **Chapter 2** This chapter contains the preliminaries, literature review and summary sections. In the literature review part, we've discussed the existing works related to our topic.
- **Chapter 3** We've discussed the requirement analysis, methodology and design of our project in this chapter and concluded the chapter with the summary section. In the methodology section, the necessary diagrams and tables are given.
- **Chapter 4** The description of the implementation of our project, the results and evaluation discussions are the contents of this chapter. The chapter ended with the summary section.
- **Chapter 5** The standards and design constraints are discussed in this chapter.
- **Chapter 6** This is the concluding chapter of this paper containing the summary of our project, and a discussion on the limitation of the project and our future plan with this project.

Chapter 2

Background

In this chapter we discuss our background study, review some of the related works and then summarize them in a table.

2.1 Introduction

We have studied and researched around 15 papers to understand the current information, previous studies and relevant histories surrounding our research topic. While reading the papers, we gathered necessary information i.e., methods they have used, Key Findings, Evaluation metrics, Datasets, etc. and merged them into a table to get an overview of our research field. Alongside our background study, we have also researched some of the applications related to our research area and presented them down below.

2.2 Literature Review

2.2.1 Covid-19 and Machine Learning related papers

Wang et. al. (2020) [28] proposed a prediction method employing rolling update mechanism with Long Short Term Memory (LSTM) to predict long-term epidemic trend of COVID-19. Instead of just passing useful information to the other cell, this rolling mechanism or optimizer updates training sample sequence every iteration. And in every iteration this technique is dependent on the present prediction output and historical data. This technique evolves the model to deal with longer prediction accurately. The performance of the model is measured

using Mean Square Error (MSE). And error rate is very low in terms of daily positive cases with data provided by an ArcGIS platform published by Dong et. al. (2020) [29].

Vinay Kumar Reddy and Chimmula, Lei Zhang (2020) [30] presented a deep learning recurrent neural network (RNN) approach using Long Short Term Memory (LSTM) architecture to predict the outbreak of COVID-19 in Canada. Long Short Term Memory (LSTM) handles the deep learning network vanishing gradient problem by self-looped gates and dynamic parameter changing with adjusted the time scale. They also done sensitive analysis of important features. The result showed 93.4% accuracy while Root Mean Square Error (RMSE) is 34.83. The dataset is collected form real time database provide by Johns Hopkins University and Canadian Health authority.
Sayantani Basu, Roy H. Campbell (2020) [31] proposed a special recurrent neural network model named Long Short Term Memory (LSTM) to predict the cumulative COVID-19 cases and deaths. LSTM has the capabilities to capturing the long-term dependencies. By tuning lookback, lookahead parameter LSTM can give reasonable results. The performance is measured using Root Mean Squared Error (RMSE). It is addressed that lower the values of focused parameters the lower the RMSE. This proposed model shows efficiency in predicting deaths and claimed some assumption based on their results that counties In United States mitigation measures failed or not. Dataset contains categorical and numerical values, taken form Jhon Hopkins University repository.
Nezir & Gokhan (2020) [32] proposed a new hybrid model, which determines the factors and analyses effectiveness of countries against Covid-19. The model they suggested comprised with three processes. First of all, clustering analysis were made using K-means and Hierarchic clustering method, to optimize the number of clusters. Then, a novel DEA model named Weighted Stochastic Imprecise Data Envelopment Analysis (WSIDEA) were used to measure efficiency analysis of countries. Finally, parameters analyzing and determining necessary factors identification were accomplished with decision tree & random forest algorithms. The dataset used in this study contains data of 142 countries and they collected it from www.Kaggle.com. After collecting the data, they processed the dataset with techniques similar to Normalization, Imputation etc. The clustering analysis showed that the optimum number of clusters for 142 countries is three. Out of 142 countries 20 countries were found fully effective, 52 countries

were found 90% effective and the rest of the countries were partially effective. Then, in the efficiency analysis, agglomerative coefficient (AC) was used to measure the efficiency of clusters. Finally, parameters analyzing and identifying the variables that form the efficiency levels of the countries were analyzed with decision tree and random forest algorithms and figured out the best model. Comparison of the models was made using three criteria for regression analysis: R2, root mean square error (RMSE) and mean absolute errors (MAE). In conclusion, this study shows us comparison of different ML methods based on performance, as well as independent & dependent variables identifying and most importantly which variables directly affect the prediction

D.P. Kavadi, R. Patan and M. Ramachandran et al.(2020) [33] proposed a partial derivative regression and nonlinear machine learning (PDR-NML) method for global pandemic prediction of COVID-19. Firstly, They used a Progressive Partial Derivative Linear Regression (PPDLR) model to search for the best parameters in the dataset. Next, a Nonlinear Global Pandemic Machine Learning model (NGP-MLP) was applied to the normalized features for making accurate predictions. Finally, outbreak predictions were made by applying Kuhn–Tucker conditions to the support vector values. They used "COVID-19 Indian database", Kaggle, containing the new confirmed daily case reports for the states and union territories of India. They found out prediction accuracy improvement was found using the proposed PDR-NML method with the application of the NGP-MLP algorithm. This step in turn improves the prediction accuracy using the PDR-NML method by 2% compared to the linear regression and 3% compared to the AI-based method. AI algorithms and correlative inferences do not scale globally to big data problems, and the correlation evaluation still depends on point estimates of limit variables. These problems solved by proposing a PPDLR model and acquiring coarse- grained predictive grouping using a Nonlinear Global Pandemic.

Cafer Mert Yes¸ ilkanat(2020): [34] proposed a method to evaluate the performance of the Random Forest (RF) machine learning algorithm was investigated in estimating the near future case numbers for 190 countries in the world and it is mapped in comparison with actual confirmed cases results. They divided into 3 main sub-datasets: training sub-data, testing sub-data (interpolation data) and estimating sub-data (extrapolation data) for the random forest model. At

the end of the study, it has been found that R2 values for testing sub-data of RF model estimates range between 0.843 and 0.995 (average R 2 = 0.959), and RMSE values between 141.76 and 526.18 (mean RMSE = 259.38); and that R2 values for estimating sub-data range between 0.690 and 0.968 (mean R2 = 0.914), and RMSE values between 549.73 and 2500.79 (mean RMSE = 909.37). These results show that the random forest machine learning algorithm performs well in estimating the number of cases for the near future in case of an epidemic like Novel Coronavirus, which outbreaks suddenly and spreads rapidly. The COVID-19 data repository used in the study was obtained from the Johns Hopkins University Center for Systems Science and Engineering (JHU CSSE). The entire study was conducted in the R programming.

Abdelhafid Zeroual , Fouzi Harrouc , Abdelkader Dairi d, Ying Sun (2020) [35] proposed the usage of 5 different deep learning methods, in order to stave off the exponentially increasing effects of the novel coronavirus (COVID-19). These deep learning methods include; Recurrent Neural Network (RNN), Long short term memory (LSTM), Bidirectional LSTM (BiLSTM), Gated recurrent units (GRUs) and Variational AutoEncoder (VAE), all of which are used in a comparative study in order to verify the eligibility of these algorithms in determining the current and predicted COVID-19 cases with a small data-set from 6 countries; namely, USA, Australia, Spain, France, Italy and China. With the intention of being better off at determining the rate of COVID-19 around the world, these deep learning methods and their studies have shown that the Variational AutoEncoder (VAE) to be the superior algorithm in order to calculate the intended findings, in regards to its performance.

Peipei Wanga , Xinqi Zheng, Jiayang Li, Bangren Zhu (2020) [36] proposed an experiment that delves into the trend of the outbreak of the novel Coronavirus (COVID-19). With how fast this virus has spread across the world, the last epidemiological information used for this study was taken from June 16, 2020, at exactly 3:33pm, and it showed 8,044,683 confirmed cases of COVID-19. The World Health Organization (WHO) stated that the US, UK, India, Brazil and Russia had the most number of confirmed cases as of June 16th. This information as of June 16th, was assimilated into the Logistics model while made to fall in line with the epidemic's trend. The information was then fed into a machine learning based time series prediction model (known as the FbProphet model) in order to antici-

pate the virus's trend over time. The FbProphet model spat out the number 14.12 million (approximately), to have been infected by late October cumulatively.

Talha Burak Alakus et. al. (2020) [37] developed and applied six Deep Learning(DL) models namely Artificial Neural Networks(ANN), Convolutional Neural Networks(CNN), Long-Short Term Memory (LSTM), Recurrent Neural Networks (RNN), CNNLSTM, and CNNRNN to predict the COVID-19 infection instead of using traditional Machine Learning(ML) models. Recall, Area Under Curve (AUC) score, Accuracy, Precision, and F1-score evaluation metrics were used to evaluate the models' performances. Moreover, the 10-fold cross-validation and the train-test split approaches were used to split the dataset for validation and testing purposes. Results showed that all the developed DL models are at least 84% accurate and 86% precise, have at least 90% of F1-Score, 92% Recall, and 52% AUC scores. DL models have many layers, which helps the models to produce the best results for larger datasets with more computational complexity. The dataset consists of 18 laboratory findings of 600 patients from the Hospital Israelita Albert Einstein, Sao Paulo, Brazil. Usually, DL models perform better with large datasets with many instances. The six DL models produced excellent results though the dataset is much smaller and has a few instances, which is one of the findings of this paper and that is that DL models can perform well with smaller datasets as well. Another key finding of this paper would be that DL models can be used to identify the COVID-19 infection without using X-ray or CT-scan images. The small and imbalanced dataset were the limitations of this work, as the models' performances can be increased with larger datasets.

Milind Yadav et. al. (2020) [38] proposed a novel Support Vector Machine for Regression (SVR) model to predict the spread and the final result of the virus, and to analyze the growth rates and the transmission rate of the virus, and the types of mitigations and their effectiveness in preventing the loss caused by the virus. And they presented Pearson's Correlation method to correlate the spread of the COVID-19 coronavirus with different weather conditions. They also used two commonly used existing regression models, the Linear Regression and the Polynomial Regression methods to compare the performance of the proposed model with them. The proposed SVR model produced the best accuracy results in most cases that made it the best performing model among the three models. The SVR model uses support vectors and finds the optimal hyperplane us-

ing them that helps the model to learn the pattern of the data more accurately. Moreover, the kernel functions used in SVR increase the computational speed of the model. The authors claimed that several advantages are gained by doing these tasks. Firstly, the spread level of the virus and the effective mitigations are identified from the first two tasks, which could help decision-makers to make advantageous decisions. Then from the third and the fourth tasks, the number of recovered patients and the areas that have fast spread rates can be found, which could help to focus the areas more that needs to be. And from the final task, a clear understanding of the correlation between different weather conditions and the spread of the virus is found, which is helpful to know whether to avoid certain weather conditions or not. The authors considered the number of positive cases, recovered cases, and death cases from 22-01-2020 to 24-04-2020 in different regions, dates, and region names as attributes for the dataset. And for the fifth task, the daily count of infected people, temperature, humidity, and the wind speed of that day were considered as the dataset attributes.

Abdelhafid Zeroua (2020) [35] presented five Deep Learning (DL) models namely Recurrent Neural Networks (RNN), Long Short-term Memory (LSTM), Bi-directional LSTM (BiLSTM), Gated Recurrent Units (GRU), Variational AutoEncoder (VAE) for time-series forecasting for 17 days of COVID-19 cases based on the daily confirmed and recovered cases of Italy, Spain, France, China, USA, and Australia using a small time-series dataset, and to identify the best performing DL model among them. The authors used Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), Mean Absolute error (MAE), explained variance (EV), and Root Mean Squared Log Error (RMSLE) as the evaluation metrics to evaluate the models' performances. Results suggest that the VAE model achieved the best performance regarding the forecast of future COVID-19 cases. As per the authors, this is the first research work that used the VAE model for forecasting COVID-19 cases. So, this is one of the key findings of this work that the VAE model can be considered for forecasting COVID-19 cases as well as the other DL models as the other models also showed promising results. The VAE model has both an encoder and a decoder that are used in encoding and decoding the original data before and after training the model to get better results. The process of encoding and decoding data from the original data is made more efficient with the help of the reconstruction loss. These can be the reasons for the model to produce better results than the other models used. The datasets used in this re-

search work consist of the COVID-19 cases of 148 days (from January 22, 2020, to June 17, 2020) of the respective countries, and they are available publicly in the Github repository of the Center for Systems Science and Engineering (CSSE) at Johns Hopkins University.

2.2.2 Exploratory Data Analysis (EDA) related papers

Jayesh and Shilpa [39] Sreedharan: Analyzed the Covid-19 Cases in Kerala. The analysis was completed over three different phases, each running one after another, commencing from the 30th of January in 2020, and finishing on the 31st of May, 2020. The first phase ran from January 30th till March 9th, the second phase from March 10th till the 8th of May and the final phase from the 9th of May till the 31st of May. As the first phase begins on the 30th of January, this is where the analysis initiates, with the first reported case of COVID-19. 3 patients or cases were reported during this initial phase, where no one else was affected, and all three patients recovered and were discharged by the 20th of February. Following this, no more reports of this virus were made till the 9th of March. With the second phase came the highest number of reports of COVID-19, where Kerala had hundreds of them. April 11th saw 268 active cases, but the numbers were seen to be decreasing afterwards. However, with such a large increase in cases, the recovery rate was blemishing. With 61 people recovering from the disease on May 4th, only 16 active cases remained on the 9th of May. May actually began with good news, as there weren't new cases to be found, and Kerala was seen to have flattened the curve for the disease. Phase 1 did not bring about any deaths, but phase 2 had three deaths and six deaths were seen in the third phase. However, the fatalities were known to have other diseases as well. The third phase started on the 9th of May, and the number of cases kept increasing. The MATLAB software was used in order to analyse the data, and all this data was sourced from various places.

Arora et al. (2020)[40] presented analyzation of South Korea's Covid-19 data employing Exploratory Data Analysis (EDA). Exploratory Data Analysis (EDA) can tell us more about data without implementing the traditional tasks. They tried to analyze the data with visualizations. This analysis helped them to find some important aspects. This study showed that how the number of confirmed, recovered cases increases, on the other hand, the number of active, deceased cases

decreases on daily basis. They also found that Daegu and Gyeongsangbuk-do is the largest cluster of infection in South Korea having 62.8% of total cases. Male patients are 59.5% and females are 40.5% of the total cases till May 5, 2020, in South Korea. This data showed that the death rate of patients aged above 60 is 92.2% with a fatality rate of 2.36%. Based on this data, South Korea has 0.11% of the total Covid-19 cases worldwide. This significant result stated that how important is technology in terms of fight this kind of global pandemic. Free diagnostics, data-driven alert system, Quarantine Inspection Procedures (QIP), Epidemiological Investigation Support System are the main key factors to reduce the infection in South Korea. The authors used OriginPro 2016 software (Version 9.3.226) for visualizations and collected data from many sources such as Korea Centers for Disease Control and Prevention (KCDC), the Ministry of Health and Welfare of South Korea (MOHW), John Hopkins GitHub repository and Coronavirus Worldometer.

Joanita DSouza Senthil Velan S (2020) [41] applied Exploratory Data Analysis (EDA) technique to COVID-19 datasets to visualize the data and extract some important information from the data. The authors used Jupyter Notebook as the environment, Python with libraries Pandas, Seaborn as the programming language for this work, and used the dataset from a GitHub repository. Different graphs were plotted for visualizing the data, namely bar charts, scatter plots, line plots, and correlation matrix. The authors represented the number of instances of each important feature in a table. **Table-2.1** shows the instance count of each

No.	Feature	Instance Count
1	TotalHospitalizedPatients	23635
2	Recovered	63120
3	Deaths	26384
4	TotalPositiveCases	195351
5	TestsPerformed	1186526000000

Table 2.1: Total instances in each feature

column. Then they visualized the region names arranging them in descending order with respect to total positive cases using a bar chart. The top three regions with the most positive cases were Lombardia, Veneto, and Emilia-Romagna ac-

cording to the bar chart they plotted. After that, they plotted a correlation matrix representing the correlation among the features. **Table-2.2** shows the results

No.	Variable on the X-axis	Variable on the Y-axis	Correlation Analysis
1	Total Positive Cases	Home Confinement	Strong Correlation of coefficient of 0.97
2	Hospitalized Patients	Total Positive Cases	Strong Correlation of coefficient of 0.94
3	Total Positive Cases	Deaths	Strong Correlation of coefficient of 0.98
4	Hospitalized Patients	Tests Performed	Weaker Correlation of coefficient of 0.75
5	Tests Performed	Deaths	Weaker Correlation of coefficient of 0.72
6	New Positive Cases	Recovered	Weaker Correlation of coefficient of 0.62

Table 2.2: Correlation of variables

they obtained from the correlation matrix. They also found that the features 'TotalHospitaizedPatients' and 'TotalPositiveCases' have a strong correlation coefficient of 0.93, and the features 'NewPositiveCases' and 'IntensiveCareUnit' have a strong correlation coefficient of 0.91 from two different plots. However, the dataset used in this work contains old data, and thus, the results obtained may be different from later works as per the authors. Otherwise, this paper extracted some important information from the dataset by applying EDA techniques quite well.

Samrat K. Dey, Md. Mahbubur Rahman, Umme R. Siddiqi and Arpita Howlader [42] proposed a model in the hopes of being able to achieve strategies that deter the rate at which the novel coronavirus Covid-19 has spread throughout the world. They did this by using the datasets provided by various institutions who are on the front line when it comes to dealing with this pandemic (John Hopkins University, World Health Organization, National Health Commission, etc). These data sets involved the usage of various types of information, including; confirmed cases of Covid-19, recovered cases of Covid-19 and the deaths caused by Covid-19. The Visualization models provided allowed for the researchers to understand more about the virus and the pneumonia-like effects it causes on the human species. It provided an interactive and user friendly canvas by which raw data could be understood comprehensively and conveniently. With the help of the data sets provided, we can look through the important statistics involved

with such pandemics and viruses, in order to help researchers and scholars attempt to fight these cases with greater knowledge and understanding of the situation.

In this study, Amobi et al. [43] used exploratory data analysis to identify pattern and nature of data and used supervised machine learning with Empirical Bayesian Kriging (EBK) to find correlation of infectious disease with Covid-19 its spread in sub-saharan Africa. They used two non-graphical (tabular) EDA technique and some graphical EDA techniques. Cross Validation (cv), minimum Bayesian Information Criterion (minBIC) adaptive (3 LASSO techniques) were used to select variables, additionaly to show that the adaptive lasso model has the smallest mean square error (MSE), 2.72 *10-28 and the largest R-squared (100%) in the testing datasets are presnted in the first table. In the second table, Selected variables were included for inference model using Cross-Fit partialing out technique and this table shows us LASSO regression for inference could statistically significantly predict coronavirus disease outbreak in SSA. The Histogram combined with line graph (Dual Axis Chart) is used for showing the contribution of countries with most Covid-19 outbreak in SSA. Next, A Matrix plot is used by authors to show correlation of seventeen continuous variables and COVID-19 disease, along with Spearman's rank-order correlation was run to determine the relationship between the two categorical variables and COVID-19. A positive correlation between transmission classification and COVID-19 disease, (rs (37) = 0.411, p = 0.009) has been found and marked in these plots. After the above mentioned figure, Begining day to end day's status of infection rate via local transmission imported cases are plotted in a quadrant chart. Cartesian graph is used to demonstrate steady and consistent growth in the total number of COVID-19 cases reported across SSA between a week of March. A multi line scatter plot is used to display the upward trend of trend cluster anlysis performed upon the countries of SSA and five clusters were formed. It also shows, South Africa is the country with highest numbers of outbreak. The global spatial autocorrelation Moran´s I statistics confirmed the random distribution pattern of coronavirus outbreak among SSA countries in Figure 6. Interpolated maps visualizes that Covid-19 is increasing per day and spreading outwards. In the last figure, coronavirus deaths tend to be concentrated in western African regions relative to central and southern African regions.

2.3 Summary

	Paper Title	Method(s)	Key Finding(s)	Evaluation Metrics	Dataset(s)
1	Comparison of deep learning approaches to predict COVID-19 infection	ANN, CNN, LSTM, RNN, CNNLSTM and CNNRNN	Usability of DL models over ML models and Laboratory findings over X-ray and CT-scan images	Recall, AUC, F1-score, Accuracy and Precision	Laboratory findings of patients from Hospital Israelita Albert Einstein
2	Analysis on novel coronavirus (COVID-19) using Machine Learning methods	SVR and Pearson's Correlation method	A novel model to predict future state of the virus and influence of weather on spreading of the virus	Coefficient of Determination, Accuracy and Associativity	Active cases, Death cases, Region names and weather info date-wise
3	Time series prediction for the epidemic trends of COVID-19 using the improved LSTM deep learning method: Case studies in Russia, Peru and Iran	Long Short Term Memory (LSTM) embedded with rolling update mechanism	Long term epidemic prediction (Both decline and rising curves)	Mean Squared Error (MSE) Russia - 1.425% , Peru-3.1% and Iran - -0.1925%	John Hopkins University's public ArcGIS platform

	Paper Title	Method(s)	Key Finding(s)	Evaluation Metrics	Dataset(s)
4	Time series forecasting of COVID-19 transmission in Canada using LSTM networks and Long Short Term Memory (LSTM)	Long Short Term Memory (LSTM)	Real time data fitting, solved gradient vanishing problem, sensitivity analysis of features	Canada(Rot mean squared error-34.83 And accuracy-93.4%)	Canadian Health Authority and Johns Hopkins University
5	COVID-19 Future Forecasting Using Supervised Machine Learning Models.	LR, LESSO, SVM and ES	ES performs best on time-series dataset having very limited entries and R squre with 98%	R square, Mean Square	Canadian Health Authority and Johns Hopkins University
6	COVID-19 Spatio-temporal estimation of the daily cases of COVID-19 in worldwide using random forest machine learning algorithm.	Random Forest (RF).	RF models perform well on epidemic predictive problems like COVID-19 in the near future	R2 values of RF between 0.843 and 0.995, and RMSE values between 141.76 and 526.18.	Johns Hopkins University's public ArcGIS platform

	Paper Title	Method(s)	Key Finding(s)	Evaluation Metrics	Dataset(s)
7	Deep Learning Methods for Forecasting COVID-19 time-series data: A Comparative Study	RNN, LSTM, BiLSTM, GRUs, VAE	VAE model out-performed the other 4 models by providing good forecasting performance with lower RMSE, MAE, MAPE, EV and RMSLE.	MAPE values of VAE for Italy 5.90%, Spain 2.19%, France 1.88%, China 0.128%, Australia 0.236%, and USA 2.04%	John Hopkins University
8	Prediction of Epidemic Trends in COVID-19 with Logistics Model and Machine Learning Techniques.	Logistic Model	Short term forecast of machine learnings are good but the long term are inferior. Hybrid Logistic and Prophet model has a valuable advantage in terms of forecasting the epidemic trend.	Nonlinear least square method	John Hopkins University

	Paper Title	Method(s)	Key Finding(s)	Evaluation Metrics	Dataset(s)
9	Assessing countries' performances against COVID-19 via WSIDEA and machine learning algorithms	WSIDEA, Decision Tree, Random Forest, K means , Hierarchic clustering	Effectiveness of countries against Covid-19, Parameters which has a correlation with Covid-19 Prediction.	Agglomerative co-efficient, R**2,RMSE,MAE	kaggle
10	Partial derivative Nonlinear Global Pandemic Machine Learning prediction of COVID 19	partial derivative regression and nonlinear machine learning	Prediction time is lower than AI – algorithms and Linear Regression. Scale globally to big data problems.	Prediction time is lower than AI – algorithms and Linear Regression. Scale globally to big data problems.	COVID-19 Indian database
11	Current perspective of COVID-19 spread across South Korea: exploratory data analysis and containment of the pandemic	Exploratory Data Analysis (EDA)	Without help of traditional techniques Exploratory Data Analysis (EDA) can help to extract important aspects. Data visualization can also extract features.	Number of confirmed, active, deceased cases.	Korea Centers for Disease Control and Prevention (KCDC), Ministry of Health and Welfare of South Korea (MOHW), John Hopkins GitHub repository and Coronavirus Worldometer.

	Paper Title	Method(s)	Key Finding(s)	Evaluation Metrics	Dataset(s)
12	Analysing the Covid-19 Cases in Kerala: a Visual Exploratory Data Analysis Approach	Exploratory Data Analysis (EDA)	Exploratory Data Analysis (EDA) results concludes that the model raised by Kerala for fighting against Covid-19 can be considered as a benchmark for how the public health department can be utilised properly.	New cases, death rates, testing, welfare measures were considered for the analysis.	World Health Organisation, John Hopkins, Tweets and Facebook corona updates.

Table 2.3: Summary Table

Chapter 3

Methodology

3.1 Support Vector Reggression (SVR)

Support Vector Regression (SVR) models are implemented by following the same principles as Support Vector Machine (SVM). But instead of classifying the instances as SVM does, SVR is a regression model that plots a line or curve to predict continuous values. The curve formulation function is shown in **Eqn-3.1**.

$$f(x,w) = \sum_{i=1}^{M} w_i x^i + b \tag{3.1}$$

Here, M is the degree of the polynomial used to approximate the function.

In Simple Linear Regression, the goal is to minimize the Ordinary Least Error (OLS). **Eqn-3.2** shows the function that is used to measure the OLS values.

$$MIN \sum_{i=1}^{M} (y - w_i x_i)^2 \tag{3.2}$$

Whereas SVR focuses on minimizing the norm of the feature vector w, meaning that, it tries to find the optimized curve or line by minimizing w. The function shown in **Eqn-3.3** is used by SVR to find the norm of the feature vector.

$$MIN_w 1/2||w||^2 \tag{3.3}$$

However, SVR is given some sensitivity to the error with the help of support vector lines or curves. Two boundary lines or curves are formulated on both sides of the learning function at a distance of ε. The two border-lines act like a tube

called ε-tube that gives the model some sensitivity to the errors, meaning the points inside the ε-tube are not considered for error calculation even though the learning function doesn't properly fit those points. The constraint of measuring error is calculated using the function shown in **Eqn-3.4**.

$$|y_i - w_i\ x_i| \leq \varepsilon \tag{3.4}$$

Moreover, the slack variables ξ^* and ξ are added to the model as hyper-parameters. These two variables calculate the distance of each point fallen outside the ε-tube. So, the updated feature vector minimizing function and the constraint function are as shown in **Eqn-3.5** and **Eqn-3.6** respectively.

$$MIN\ 1/2\ ||w||^2 + C\sum_{i=1}^{N} \xi^* + \xi \tag{3.5}$$

$$|y_i - w_i\ x_i| \leq \varepsilon + |\xi_i| \tag{3.6}$$

SVR can work with high-dimensional datasets with significantly lesser time than other regression models with the help of kernel functions. There are different types of kernel functions are used, however, **Eqn-3.7** shows the general function of the kernel funcitons.

$$k(x_i,\ x) = \phi(x_i) * \phi(x) \tag{3.7}$$

Here, $\phi()$ is the transformation function that transforms the feature space to kernel space and thus, it reduces the time of calculating the dot product of two higher-dimensional vectors significantly.

3.2 Holt-Winters Triple Exponential Smoothing

Another one of Holt's methods is the Triple Exponential Smoothing process. This logs and compares the changes between trends and seasonal changes, allowing for Holt to use a method to capture seasonality. The noteworthy factors in the relevant equations are: l t (for level), b t (for trend) and s t (for seasonal component). The smoothing parameters are α , β^* and γ.

Seasonality takes into account the recurrence in every t period for the behavior of time-series data. Seasons represent the period of time before this recurrence repeats. This method consists of two versions: additive and multiplicative.

3.2.1 Holt-Winters Additive Method

$$Level(l_t) = \alpha(y_t - s_{t-m}) + (1-\alpha)(l_{t-1} + b_{t-1}) \tag{3.8}$$

$$Growth(b_t) = \beta^*(l_t - l_{t-1}) + (1-\beta^*)b_{t-1} \tag{3.9}$$

$$Seasonal(s_t) = \gamma(y_t - l_{t-1} - b_{t-1}) + (1-\gamma)s_{t-m} \tag{3.10}$$

$$Forecast(\hat{y}_{t+h|t}) = l_t + hb_t + s_{t-m+h_m^+} \tag{3.11}$$

where m is the length of seasonality in months, l_t represents the level of the series b_t denotes growth, s_t is the seasonal component, $\hat{y}_{t+h|t}$ is the forecast for h periods ahead, and $h_m^+ = [(h-1)modm]+1$. The parameters (α,β^* and γ) are usually restricted to lie between 0 and 1.

$$s_t = \gamma^*(y_t - l_t) + (1-\gamma^*)s_{t-m} \tag{3.12}$$

if we substitute for l_t from the seasonal smoothing equation (3.10) for the equation for the level of the component form we get

$$s_t = \gamma^*(1-\alpha)(y_t - l_{t-1} - b_{t-1}) + [1-\gamma^*(1-\alpha)]s_{t-m} \tag{3.13}$$

which is identical to the smoothing equation for the seasonal component $\gamma = \gamma^*$ (1 - γ). The Error correction term of the smoothing equation is given by

$$l_t = l_{t-1} + b_{t-1} + \alpha e_t \tag{3.14}$$

$$b_t = b_{t-1} + \alpha\beta e_t \tag{3.15}$$

$$s_t = s_{t-m} + \alpha e_t \tag{3.16}$$

where $e_t = y_t$ (l_t 1 + b_t 1 + s_t m) = y_t $\hat{y}_{t|t-1}$ are one-step training forecast errors.

3.2.2 Holt-Winters Multiplicative Method

$$Level(l_t) = \alpha\frac{y_t}{s_{t-m}} + (1-\alpha)(l_{t-1} + b_{t-1})t - 1 \tag{3.17}$$

$$Growth(b_t) = \beta^*(l_t - l_{t-1}) + (1-\beta^*)b_{t-1} \tag{3.18}$$

$$Seasonal(s_t) = \gamma\frac{y_t}{l_{t-1}+b_{t-1}} + (1-\gamma)s_{t-m} \tag{3.19}$$

$$Forecast(\hat{y}_{t+h|t}) = (l_t + hb_t)s_{t-m} + h_m^+ \tag{3.20}$$

See (3.20). and the error correction component is given by;

$$l_t = l_{t-1} + b_{t-1} + \alpha\frac{e_t}{s_{t-m}} \tag{3.21}$$

$$b_t = \beta_{t-1} + \alpha\beta^*\frac{e_t}{s_{t-m}} \tag{3.22}$$

$$s_t = s_t + \gamma\frac{e_t}{l_{t-1}+b_{t-1}} \tag{3.23}$$

where $e_t = y_t - (l_{t-1} + b_{t-1})s_{t-m}$ and α , β , and γ are constants that must be estimated in such a way that the MASE of the error is minimized.

3.3 Polynomial Regression

Polynomial regression is a type of linear regression that involves two variables, one independent (x) and one dependent (y), where their relationship formulates as an nth order polynomial. It uses both x and y, where x is used to determine y. X is the set of regular attributes, and y is the label attribute. E(y|x) is the concluding relationship that is created in polynomial regression in the nonlinear relationship between x's value and the corresponding conditional mean of y. This formulation can and has been used in order to various nonlinear phenomena, including the spreading of diseases. Although polynomial regression fits a nonlinear model to the data, as a statistical estimation problem it is linear, in the sense that the regression function E(y | x) is linear in the unknown parameters that are estimated from the data. For this reason, polynomial regression is considered to be a special case of multiple linear regression.

The goal of regression analysis is to model the expected value of a dependent variable y in terms of the value of an independent variable (or vector of inde-

pendent variables) x. In simple linear regression, the following model is used:

$$y = i_0 + (i_1 * x) \tag{3.24}$$

In this model, for each unit increase in the value of x, the conditional expectation of y increases by w1 units.

In many settings, such a linear relationship may not hold. For example, if we are modeling the yield of a chemical synthesis in terms of the temperature at which the synthesis takes place, we may find that the yield improves by increasing amounts for each unit increase in temperature. In this case, we might propose a quadratic model of the form:

$$y = i_0 + (i_1 * x^1) + (i_2 * x_2^2) \tag{3.25}$$

In this model, when the temperature is increased from x to x + 1 units, the expected yield changes by i_1 + i_2 + 2 (i_2 * x). The fact that the change in yield depends on x is what makes the relationship nonlinear. In general, we can model the expected value of y as an nth order polynomial, yielding the general polynomial regression model:

$$y = i_0 + (i_1 * x^1) + (i_2 * x_2^2 + ... + (in * xn^n)) \tag{3.26}$$

Regression is a technique used for numerical prediction. It is a statistical measure that attempts to determine the strength of the relationship between one dependent variable and a series of other changing variables known as independent variables.

3.4 Holt's Linear Trend

Double Exponential Smoothing also know as Holt's Linear Trend method. Exponential Smoothing improves the Weighted Moving Average method by containing all the previous instances and also give more importance to new ones [44]. In addition to previous Exponential Smoothing method, this method introduces a new variable to predict the trend either it is upward or downward [45]. There are total three equation used in this approach.This method is depending on previous data so, while calculation starts need to initialize the first values for these equations 3.27.

$$u_1 = y_1, v_1 = 0 \tag{3.27}$$

Here y_1 is first instance real value and u_1, v_1 is the first instance value for this methods predicting equation. The main three equations are stated below.

$$u_i = \alpha y_i + (1-\alpha)(u_{i-1} + v_{i-1}) \tag{3.28}$$

$$v_i = \beta(u_i - u_{i-1}) + (1-\beta) v_{i-1} \tag{3.29}$$

$$\widehat{y}_{i+1} = u_i + v_i \tag{3.30}$$

The equation 3.28 refers the calculation of smoothing the time series data. And the smoothing constant α has value between 0 to 1. The second equation 3.29 is for predicting the trend of the given data. Here also the trend smoothing constant β has value between 0 to 1. And finally third equation 3.30 sums the previous two equation values and it is the final predicted value. Here u_i, v_i respectively the smoothed constant and smoothed trend value for period i.

3.5 Proposed Model

The proposed model is an ensemble of the four ML models namely the Holt's Linear, the Holt-Winter's Triple Exponential Smoothing, the Polynomial Regression, and the Support Vector Regression model with weighted average approach. We've used each model independently to learn the pattern of the COVID-19 data, but not all the models performed well. So, we tried to gain better performance from the models and proposed an ensemble models consisting of those models. Based on the models' individual performances on the dataset, we assigned different weights on each of the model and took the weighted average of all the model's output as the output for each of the instances. The average was taken using the function shown in **Eqn-3.31**.

$$y_i = \frac{\sum_{j=1}^{K} y_j * w_i}{\sum_{j=1}^{K} w_i} \tag{3.31}$$

Here, y_i is the predicted value of the confirmed cases for the ith instance and y_j is the predicted value for the ith instance by each of the four ML models. In each iteration, the jth model is trained with a subset of the dataset and a value y_j is predicted as the confirmed cases for the ith instance. Randomly m instances

are selected for each model in every iteration. Finally, All the predicted values are summed up and the summed value is divided by *K*, which is the number of models used in the ensemble, to get the average value.

Chapter 4

Implementation and Results

In this Chapter we discussed briefly what is our environment setup, evaluation of methods and finally the results of our experiments.

4.1 Dataset Description

The dataset used in this project is collected from the Github Repository of the Center for Systems Science and Engineering (CSSE) at John Hopkins University. This is an integrated dataset from several datasets collected from different organizations and health departments. As the dataset is open for using in research

Serial Number	**Attributes for COVID-19**
1	SNo
2	Observation Date
3	Province/State
4	Country/Region
5	Last Update
6	Confirmed
7	Deaths
8	Recovered

Table 4.1: Primary Attributes of the Datasets

Serial Number	Attributes for COVID-19
1	Active
2	Closed
3	Days Since

Table 4.2: Secondary Attributes of the Datasets

Serial Number	Attributes for COVID-19
1	Observation Date
2	Country/Region
3	Confirmed
4	Deaths
5	Recovered
6	Active
7	Closed
8	Days Since

Table 4.3: Attributes Considered in the Project

works, we used it to get some crucial information about the COVID-19 coronavirus. The regularly updated dataset can be found in the Github Repository, but we considered data from January 22, 2020 to February 27, 2021. There exists a total of 236013 instances in the dataset with 8 features in the dataset. But we removed some of them, for example, 'SNo', 'Province/State', 'Last Update' columns were dropped as we didn't use them in our project. However, some features were added by us to the dataset using other existing features like 'Active', 'Closed' & 'Days Since'. All the features considered from the dataset are shown in **Table-4.3**. Moreover, the dataset needed some data pre-processing as there were some noisy instances with negative or null values and some features that weren't needed for this study. In the data pre-processing part, the negative values were replaced by the minimum value of that feature, the null values were replaced by 0, and as mentioned before, some features were dropped. Then the values of the features 'Confirmed', 'Deaths', 'Recovered', 'Active' and 'Closed' were nor-

malized using **Eqn-4.1** for the models to produce better results.

$$x' = \frac{x_i - x_{min}}{x_{max} - x_{min}} \tag{4.1}$$

4.2 Exploratory Data Analysis (EDA)

4.2.1 Worldwide

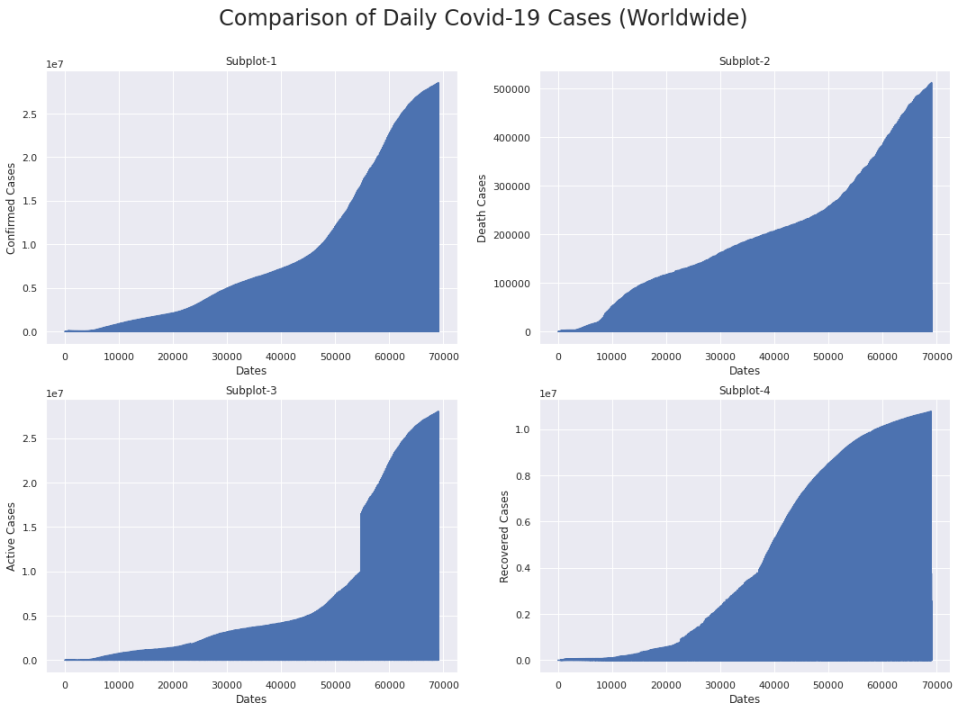


Figure 4.1: Different Covid-19 cases (Worldwide)

In the early January, 2020 World Health Organization (WHO) addressed that COVID-19 as a Public Health Emergency of International Concern. Then it was just within the Wuhan's border in China. Being a transmission based virus, in a short period of time it became global pandemic. Suddenly in March 2020 it became a global pandemic for its huge transmission all over the world. Exploratory Data Analysis (EDA) is used here to show this patterns of COVID-19. **Figure 4.1** shows the trend of confirmed, death, active and recovered cases around the world from March, 2020 till March, 2020 in four different subplots. In subplot-1, it can be seen that the number of confirmed cases has increased quite slowly be-

fore March 2020 to September, 2020. But, from then, the number of confirmed cases increased in a rapid pace till January, 2021. However, the curve seems to be flattening after that which could be an indication of the decrease in the number of confirmed cases. From subplot-2, it can be seen that in the first few months, a very few people died of the virus but, then the number of death cases increased at a slow pace till September, 2020. After that the curve becomes steeper that indicates the number of death cases have increased quite quickly from then and still rising. Then the subplot-3 shows that initially, the number of active cases increased very much slowly which is understandable because not much people were confirmed affected or were not confirmed dead at that time. However, the number of active cases increased rapidly after September, That is an indication of the fact that a large number of people are recovering from the virus. In subplot-4, the figure describes that the total number of recovered cases around the world. We can see that the number of recovered cases were very low till July, 2020. But in between the months of July and August, there is a increase in the number of recovered cases as shown in the curve.

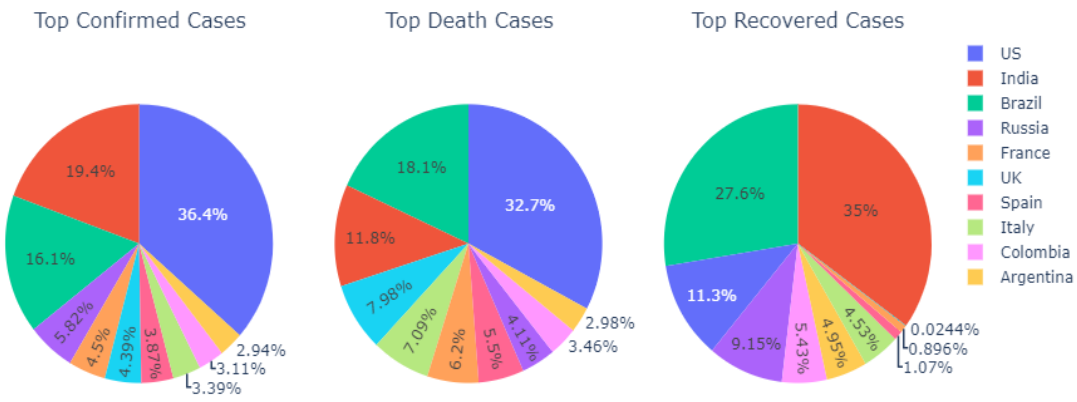


Figure 4.2: Top 10 Countries with Highest Covid-19 Cases (Pie Charts)

Not every country in the world suffered the same due to the spread of the Covid-19 coronavirus. **Figure 4.2** shows the top 10 countries in terms of the most number of confirmed, death and recovered cases using pie charts. The first pie chart on the left shows the top 10 countries with the most number of confirmed cases. United States being the top country with 36.4% of the world's total confirmed

cases, followed India with 19.4%, Brazil with 16.1%, and other countries with less than 10% of such cases. The pie chart in the middle shows the top 10 countries with the most number of death cases. United States is also the top country in terms of death cases with 12.7% of the world's total death cases. Brazil being the second with 18.1%, then India with 11.8%, and other countries with below 10% of the world's total death cases. And the third pie chart on the right shows the top 10 countries with the most number of recovered cases. India has the most recovered cases which is 35% of the world's total recovered cases, followed by Brazil with 27.6%, US with 11.3%, and the rest being below 10%.

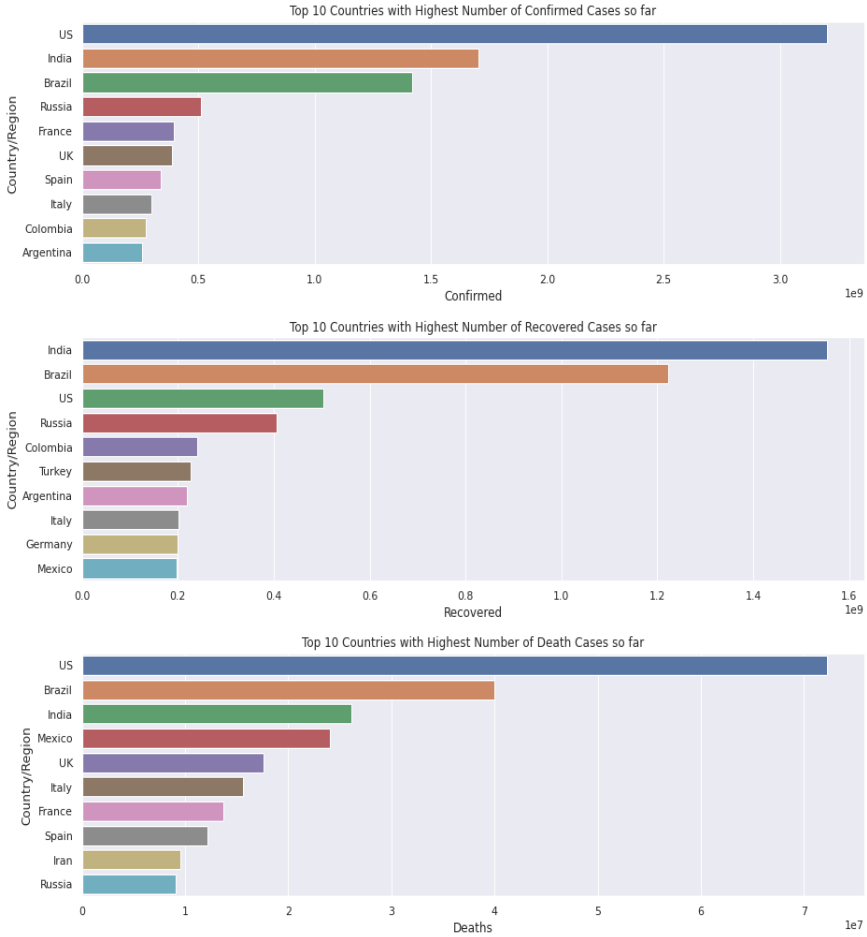


Figure 4.3: Top 10 Countries with Highest Covid-19 Cases (Horizontal Bars)

Different Covid-19 cases namely confirmed, recovered and death cases are also

shown using horizontal bars in **Figure 4.3** which confirms the results got from the pie charts shown in **Figure 4.3**. These horizontal bar plots also shows that the top countries affected by the Covid-19 coronavirus are the countries with large number of population like United States, India, Brazil, Russia, etc. By observing the pie charts, some key points is found. Firstly, countries with large areas and large population have been on top in term of confirmed cases. Secondly, as those countries have the most number of people affected by the virus, so understandably, they also have most number of people died of the virus and recovered from the virus. The steps taken by those countries' governments to prevent virus from spreading in their countries can be researched for to prepare for future consequences.

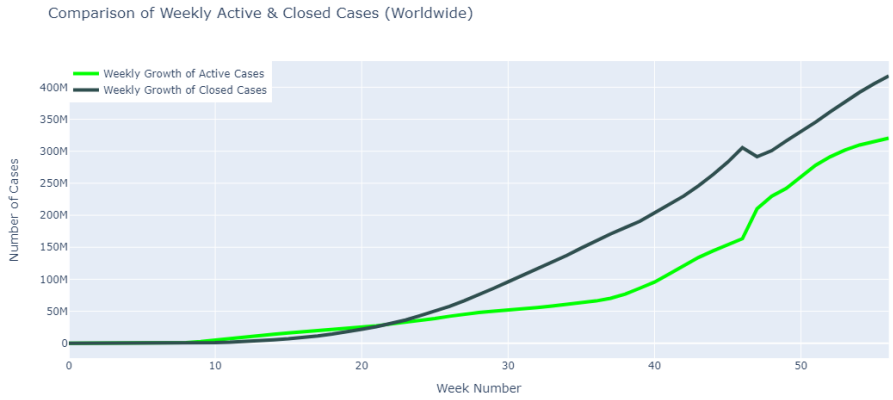


Figure 4.4: Weekly changes in active and closed cases (Worldwide)

Active cases represents the number of patients suffering from the virus while closed cases refers to the total number of patients that are recovered from the virus or have died of the virus. The weekly changes in worldwide active and closed cases are shown in **Figure 4.4**. The black curve representing the weekly closed cases and the lime one representing the weekly closed cases worldwide. Both the curves indicate exponential increase. The active cases are calculated by subtracting the number of recovered and death cases from the confirmed cases in each week. And the closed cases are the summation of recovered and death cases. It can be seen that the curve of active cases is lower than the closed cases i.e., the number of people recovered and died of the virus is more than confirmed cases in each week worldwide. But, as shown before in **Figure 4.1**, it

is known that the recovered cases are much higher than death cases, so most of the closed cases are recovered cases.

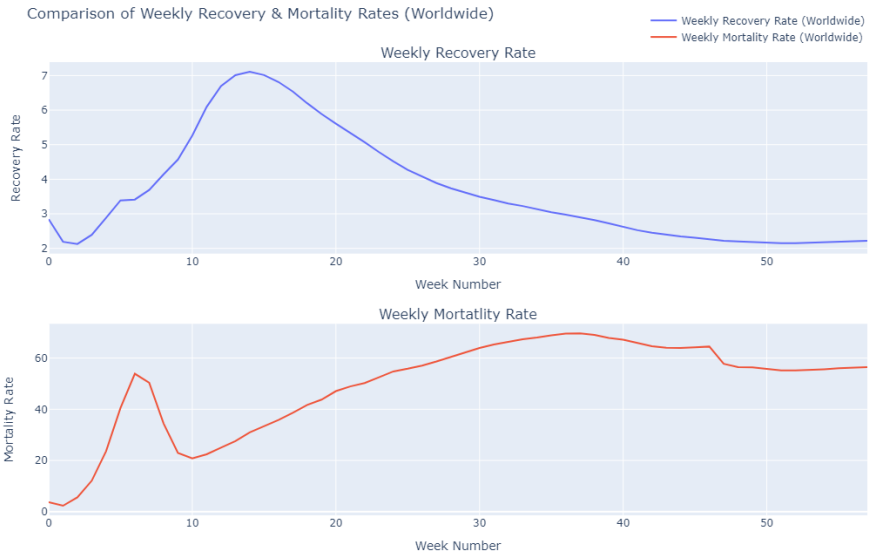


Figure 4.5: Comparison Between Weekly Recovery Rate and Mortality Rate (Worldwide)

Mortality rate describes the relative death rates in a particular population within a specified period of time. It's also useful for knowing the impact of the disease in the population, which in this case is the Covid-19 coronavirus. Similarly, the recovery rate shows the recovery rate against the number of people affected by the virus. **Figure 4.5** shows the trend of the worldwide weekly recovery and mortality rates. It shows that the curve of the recovery rate was upward till week 15, meaning that many people were recovered in that span of time. But after that the trend of the curve started going downward till week 53. This indicates the number of people recovering is getting low day by day. So, there are two possible cases for this type change in the recovery rate's curve. The first reason could be that, not many people are affected by the virus in that period and thus the number of recovered people is also less than before. And the second reason could be that, the number of death cases increased in that period of time that made the

trend of the recovery rate's curve move downward. Now, from the curve of the mortality rate, it can be seen that the mortality rate increased very quickly till week 5, but then it started decreasing and that decreasing continued for a few weeks. But then the mortality rate started increasing again as the trend of the curve is gone upward since then. This indicates that the mortality rate increased after week 10, which results in less number of recovered cases.

4.2.2 Bangladesh

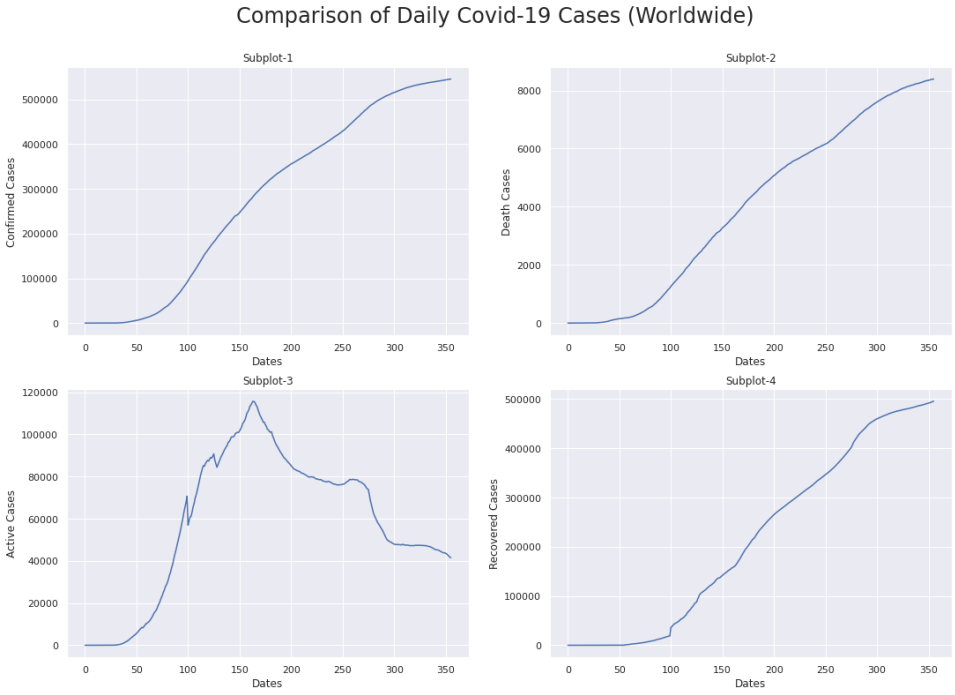


Figure 4.6: Different Covid-19 cases (Bangladesh)

A comparison among the confirmed, deaths, active and recovered cases in Bangladesh are shown in **Figure-4.6**. The number of daily confirmed cases in Bangladesh has increased quite rapidly since day 55 as per the curve shown in the subplot-1. The curves of daily death cases and recovered cases are also quite same as the curve of the confirmed cases indicating a quick increase in the number. However, the curve of active cases in Bangladesh show in the subplot-3 is much different than the others. It shows that the daily growth of active cases in Bangladesh were increasing very rapidly after day 50 and that increase in number contin-

ued till day 155. Then the number started decreasing indicating that since then, more cases were being closed. From subplot-2 and subplot-4, it can said that most of those closed cases were recovered cases, which is quite relieving for Bangladesh as of January, 2021.

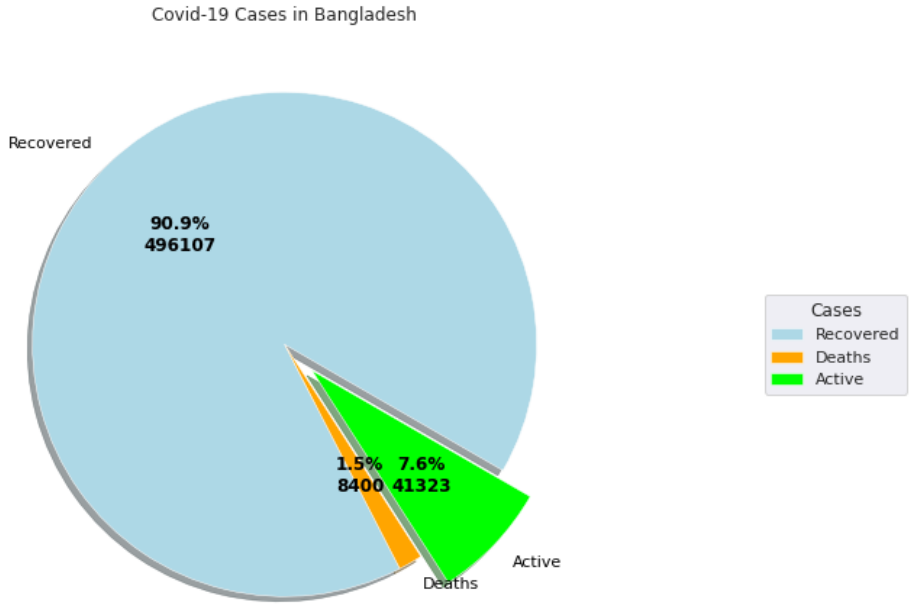


Figure 4.7: Comparison between different COVID-19 cases (Bangladesh)

Figure 4.9 shows a comparison between the recovered, deaths and active cases in Bangladesh using a pie chart. It shows that among all the Covid-19 cases, 90.9% of people have been recovered from the virus, while 1.5% among the confirmed cases have died of the virus and 7.6% people are still suffering from the virus in Bangladesh. So, from these numbers, it can be said that the recovery rate is much higher than the death rate in Bangladesh.

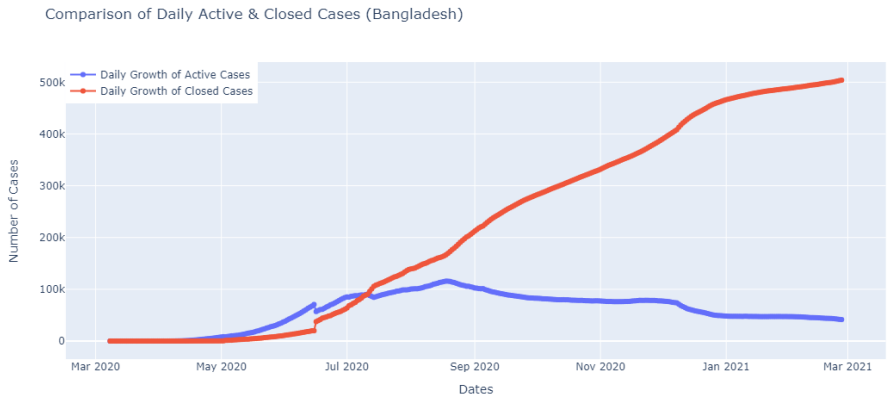


Figure 4.8: Comparison between daily Active and Closed cases (Bangladesh)

In **Figure 4.8**, two curves are shown in the same figure, the red curve representing the daily closed cases and the blue one representing the daily active cases in Bangladesh. It can be clearly seen that the curve of the daily active cases is very close to flat i.e., the number of people suffering from the virus is very low compared to the number of people affected by the virus. However, the curve of the closed cases has an exponential growth meaning that the number of daily recovered and death cases has increased very quickly. But as shown before in **Figure 4.6**, the number of daily death cases are very low compared to the number of people recovered daily in Bangladesh.

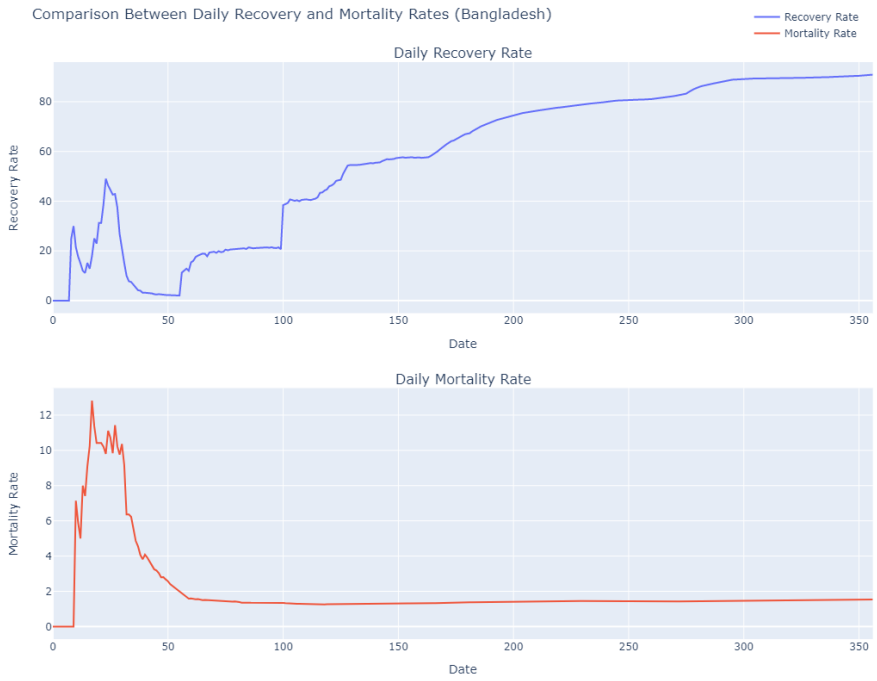


Figure 4.9: Comparison between Mortality Rate and Recovery Rate (Bangladesh)

Figure-4.9 shows the daily recovery rate and the mortality rate in Bangladesh. The curve of the recovery rate is upward trending which indicates that the number of people recovering from the virus is increasing day by day. Whereas, the curve of the mortality rate was initially upward and steeper but quickly the trend became downward and became flat. This indicates the number of people dying of the virus increased rapidly initially but in a span of a month, the number decreased very quickly as well.

4.3 Results

Four different Machine Learning models namely, Holt's Linear, Support Vector Regression, Polynomial Regression, Holt's Winter models and the proposed Ensemle Model with Weighted Average have been used on the whole dataset to learn the trend of different cases of the Covid-19 coronavirus. These five models have been used to learn the trend of each of the five features separately, those are confirmed, recovered, deaths and active cases. The performances of the models

were evaluated using the Root Mean Squared Error (RMSE), the Mean Absolute Percentage Error (MAPE) and the R^2 score. RMSE and MAPE are the measures of error produced by the models, and the R^2 Score describes how good or bad the model has performed compared to the Linear Regression model. If the value of R^2 score of a model is negative, then it indicates the model has performed worse than Linear Regression. Whereas, if the value of R^2 score is positive, then it means the model has performed better than the Linear Regression model.

Model Name	Covid-19 Cases	RMSE	MAPE	R^2 score
Holt's Linear Model (Worldwide)	Confirmed	1.08	0.04	0.96
	Recovered	0.19	0.02	0.99
	Deaths	2.08	0.08	0.70
Holt's Winter Model (Worldwide)	Confirmed	1.34	0.04	0.95
	Recovered	0.20	0.02	0.98
	Deaths	2.49	0.10	0.57
Polynomial Model (Worldwide)	Confirmed	2.74	0.11	0.77
	Recovered	0.65	0.06	0.82
	Deaths	0.99	0.03	0.93
Support Vector Regression (Worldwide)	Confirmed	2.65	0.08	0.79
	Recovered	0.58	0.06	0.86
	Deaths	1.33	0.08	0.88
The Ensemble Model (Worldwide)	Confirmed	0.94	0.03	0.97
	Recovered	0.19	0.02	0.98
	Deaths	0.38	0.02	0.99

Table 4.4: Different evaluation metric values (Worldwide)

All the values of these evaluation metrics for each of the models are shown in **Table 4.4**. The table shows that among the first four existing ML models, the Holt's Linear Model has produced the overall least RMSE and MAPE scores which are summed up to 3.94 and 0.44 respectively and also the lowest individual RMSE and MAPE scores of 019 and 0.02 for the recovered cases. Moreover, the R^2 scores of the Holt's Linear model are also positive for all the features and the total R^2

score is 2.90 which is the most among the four ML models. So, it can be said that the Holt's Linear model is the best performing model among the existing ML models used here. However, the Support Vector Regressor (SVR) model has produced the overall most RMSE and MAPE scores in total which is summed up to 7.17 and 1.49 respectively, and the most individual MAPE score of 1.27 for the active cases. The R^2 score for active cases for the model is also negative, meaning that the SVR model has failed to generate better results than the Linear Regression model for active cases. Though the values of R^2 scores are positive for other features, it could not prevent the model being the overall worst performing model among the four ML models used. On the other hand, the proposed ensemble model has generated the best results as it has the lowest RMSE value of 1.90 in total which is much less than the RMSE score of the Holt's Linear model. The propsed model also has the lowest MAPE score of 0.25 in total, which is also much less than the MAPE score of the Holt's Linear model. Among all the models, the proposed model produced the best R^2 scores except for the recovered case where the Holt's Linear model has a slightly higher R^2 score of 0.99. But, still the overall performance of the proposed ensemble model is best among all the models.

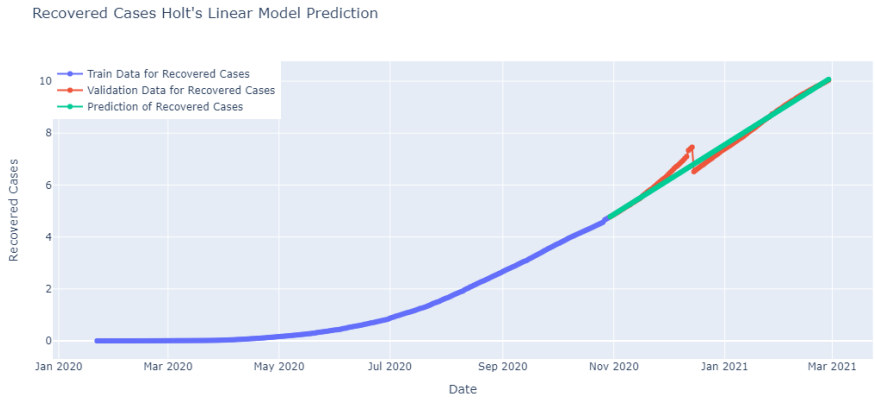


Figure 4.10: Holt's Linear for Recovered Cases (Worldwide)

Figure 4.10 verifies the results shown in **Table 4.4** above, where it can be seen the curves are very close to the actual cases for the recovered cases by the Holt's Linear model.

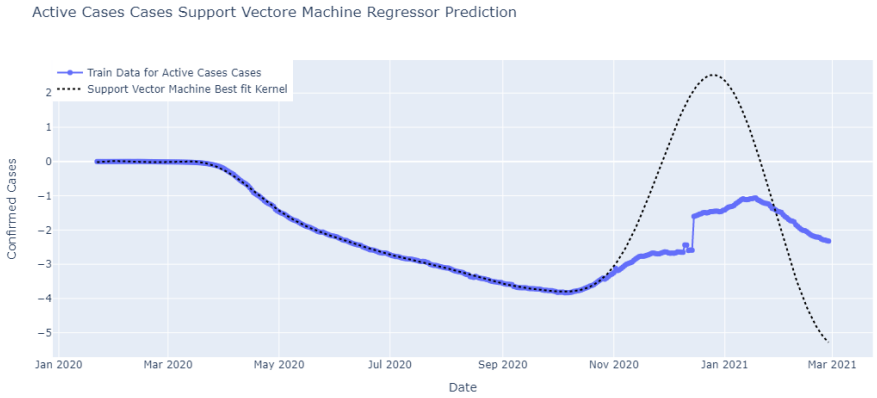


Figure 4.11: SVR for Active Cases (Worldwide)

However, **Figure 4.11** shows that the curve of the SVR model couldn't learn the pattern well of the actual cases for the active cases indicating the poor performance of the SVR model on learning the data.

Model Name	Covid-19 Cases	RMSE	MAPE	R^2 Error
Holt's Linear Model (Bangladesh)	Confirmed	0.005	0.03	0.74
	Recovered	0.003	0.03	0.84
	Deaths	0.003	0.03	0.83
Holt's Winter Model (Bangladesh)	Confirmed	0.007	0.35	0.42
	Recovered	0.003	0.03	0.83
	Deaths	0.002	0.03	0.85
Polynomial Model (Bangladesh)	Confirmed	0.02	0.09	-1.74
	Recovered	0.007	0.09	0.19
	Deaths	0.004	0.04	0.65
Support Vector Regression (Bangladesh)	Confirmed	0.06	0.39	-41.34
	Recovered	0.04	0.48	-24.38
	Deaths	0.005	0.07	0.38
The Ensemble Model (Bangladesh)	Confirmed	0.005	0.02	0.75
	Recovered	0.003	0.03	0.84
	Deaths	0.002	0.03	0.89

Table 4.5: Different evaluation metric values (Bangladesh)

The five models have also been used to learn the trend of the virus in Bangladesh. The RMSE, MAPE and R^2 score is also used to evaluate the performances of these models. **Table 4.5** shows the values for each of the models for different Covid-19 cases. The Holt's Linear model is also produced the overall best results for the cases in Bangladesh, and the SVR model is the least performing model with the highest overall RMSE and MAPE scores. And, the proposed Ensemble model has generated the best results for Bangladesh's data also. The total RMSE and MAPE scores are the least of the proposed Ensemble model, and all the values of the R^2 values are positive for all types of cases considered.

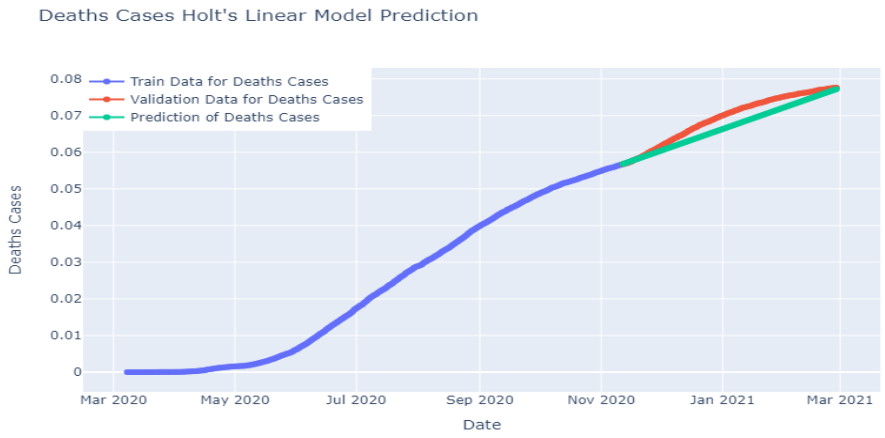


Figure 4.12: Holt's Linear Model's Curve For Death Cases (Bangladesh)

The curves in the **Figure 4.12** shows that the Holt's Linear model is also very much accurate in learning the pattern of the date-wise Covid-19 cases in Bangladesh.

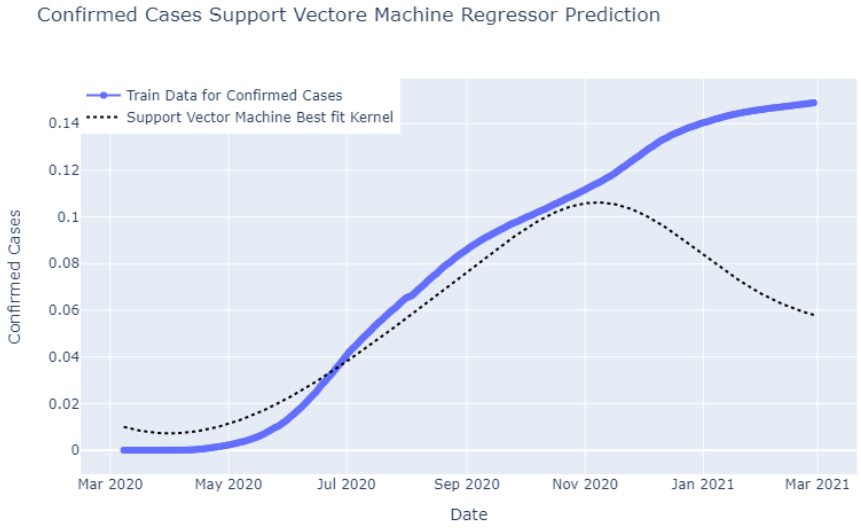


Figure 4.13: SVR Model's curve comparison (Bangladesh)

On the other hand, the SVR model is the least performing model that failed to performed well in learning the pattern of date-wise Covid-19 cases in Bangladesh.

The curve for the confirmed Covid-19 cases in Bangladesh is shown in **Figure 4.13**.

Chapter 5

Standards and Design Constraints

In this Chapter we discussed the software and hardware standards that we used during the experiment.

5.1 Compliance with the Standards

5.1.1 Software Standards

Tool Name	Standard Version
Python	3.6.9
Scikit-learn	0.21
Matplotlib	3.3.4
Seaborn	0.11.1
Plotly	4.14.3
Pandas	1.2.1
Numpy	1.20.0

Table 5.1: Standard Software Versions

Maintaining the software standard is very important to follow any project. The standard software versions shown in the **Table 5.1** are needed to be maintained

to run all the codes of our project without any hassle and to get the accurate and desired results as mentioned in **Section 4.3**. Many of the features we used of these softwares are unique to these versions only, i.e., these features are not available in the previous versions. For example, if one runs all of our project's python codes in an environment where Python version 2 is installed, then it definately won't give the desired results and it might crush the programs as well. Similarly, the versions mentioned for Matplotlib, Plotly, Pandas, and Numpy libraries should also be maintained.

5.1.2 Hardware and Runtime Environment Standards

Tool Name	Minimum Requirements
Processor	Core-i5 or above
RAM	8 GB
ROM	60 GB Free Space
GPU	4 GB

Table 5.2: Standard Software Versions

We have used Google Colaboratory as our runtime environment that is a cloud-based and a platform to write and execute codes. We did all the data pre-processing, EDA analysis tasks and also to build and train the models to extract the results in there. As Google Colaboratory is an online platform for doing such work, it is mandatory to be connected to a network with a stable internet connection. It allows the users to have free access to GPUs, so we didn't have to think about the configuration much, but if one wants to run the codes of our project in a local environment, then it's important to use a PC or Laptop with the minimum specifications mentioned in **Table 5.2**.

Chapter 6

Conclusion

In this chapter we discussed our entire project summary, limitations and future work.

6.1 Summary

In this study, we used Machine Learning (ML) to predict the future trends of the COVID-19 epidemic and extract critical features from the data. At first, we read quite a few papers related to our topic and wrote literature reviews on each of them that helped us in making the framework of the project. The dataset used in this work was collected from the Github repository the Center for Systems Science and Engineering (CSSE) at John Hopkins University, containing data from January 22, 2020 to February 27, 2021. Different Exploratory Data Analysis (EDA) techniques were used to visualize the data and necessary pre-processing tasks have been done accordingly. The data was split into train and validation sets using the train-test-split approach. Then we used four ML models, namely the Polynomial Regression, the Holt's Linear, the Holt's Winter and the Support Vector Regression models separately but couldn't get the desired results from any of them. Then, we developed an ensemble model using those four ML models to extract the pattern of the data and the results were very much promising. The models' performances were evaluated using the Root Mean Squared Error (RMSE), the Mean Absolute Percentage Error (MAPE) and the R^2 score. The four ML models couldn't produce decent performances when trained individually. The Holt's Linear model was the overall best performing model among the four ML models with the minimum RMSE, MAPE, and R^2 Score. Whereas, the Sup-

port Vector Regression model produced the highest RMSE, MAPE and R^2 Score which made it the least performing model. On the other hand, the proposed ensemble model produced the overall least RMSE, MAPE and R^2 Score.

6.2 Limitation

Training ML models require high-end machines to execute and generate results faster. Since we don't have any specialized machines to run ML models, our work speed got slowed down a bit. Moreover, the absence of a separate test dataset was a limitation of this project. To overcome that we split the dataset into train and validation datasets using the train-test-split approach. One of the limitations was the shortage of time to modify the ML models. We believe that the models can give better performances if the hyper-parameters are tuned properly. Furthermore, the dataset used in this study is missing some important features which is also a limitation of this project.

6.3 Future Work

In this study, some ML models have been used to predict the future trend of COVID-19 so far. In the future, we want to extend this work to find how effective the lockdowns and other measures taken were in controlling the spread of the virus in different countries. We want to find out the effectiveness of the vaccine to prevent the COVID-19 spread after we've collected enough data of the days of vaccination. Moreover, We want to do a detailed Exploratory Data Analysis (EDA) on Bangladesh in the future.

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